

Drive-Level Momentum in the NFL:

Most of It Is Short Fields, and the Residual Cuts the Other Way

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Abstract

Coaches, broadcasters, and fans treat NFL momentum as a within-team phenomenon, but the data, properly disaggregated, place the action between teams. Using `nflfastR` play-by-play covering 4,078 games and 85,955 drives from 2010–2024, I decompose drive-to-drive momentum into three channels: own-team defensive-stop spillover (H1), own-team offensive carryover (H2), and opponent suppression (H3). The unconditional H1 gap is large (52.2% vs. 35.6%, a 16.6-percentage-point difference). Once starting field position and game-state controls are added, along with game and possessing-team fixed effects, H1 collapses to 2.0 points and H2 turns negative (−4.2 points), reversing the broadcast prediction. The opponent-suppression channel survives all controls and is large in magnitude: scoring probability is 32.7 percentage points lower following an opponent score. A battery of eight robustness checks — flexible field-position controls, a post-touchdown-versus-post-field-goal placebo, a logit comparison, within-team-quarter permutation inference, and a rule-era split — indicates that the cross-team channel is not a linear-control artifact. The within-team channels collapse under the same controls. To my knowledge, this is the first peer-reviewed drive-level NFL test of momentum and the first to separate within-team from cross-team channels.

Keywords: momentum; hot hand; play-by-play; fixed effects; `nflfastR`; cross-team spillover

1 Introduction

The single most invoked, least defined construct in football broadcasting is momentum. The empirical case for any of it is thinner than the usage suggests, but the way the construct is described

in coaching, broadcasting, and fan discussion mixes three propositions that have different observable implications and that the data treat very differently. Pulling them apart makes a sharp contrast visible: drive-to-drive momentum, properly defined, splits into a small-or-wrong-signed within-team component and a large cross-team component, and what broadcasts call momentum is overwhelmingly a description of the latter.

The hot-hand literature originated in basketball with [Gilovich et al. \[1985\]](#)’s finding that perceived shooting streaks were a perceptual artifact: shooters’ makes and misses behave like independent Bernoulli trials. The claim held as a benchmark for three decades until [Miller and Sanjurjo \[2018\]](#) identified a selection bias in the original test. With the bias correction, modest positive streak effects appear in the original Cornell shooting data. In team sports, the question divides further: do team-level outcomes (a win, a score, a turnover) influence the team’s next outcome of the same kind, or are sequences of game events independent given the underlying quality of the teams playing?

For the NFL, to my knowledge in the peer-reviewed literature the only direct test in the *Journal of Quantitative Analysis in Sports* is [Fry and Shukairy \[2012\]](#), who examine NFL momentum at the level of full games and find no evidence that recent wins predict future wins above what team strength can explain. [Arkes and Martinez \[2011\]](#) make a parallel test for the NBA and report that, after correcting for opponent strength and home-court advantage, NBA teams that win their previous game are modestly more likely to win the next. [Steege et al. \[2021\]](#) apply an entropy test to NHL season-level streaks and reject momentum at that scale. [Lopez et al. \[2018\]](#) use a unified generalized linear mixed-model framework to compare randomness in outcomes across the four major North American sports and find that NFL outcomes are the most randomness-driven among them, consistent with weak game-level momentum signals.

Game-level tests answer a different question than the one coaches actually care about. A coach calling a timeout in the third quarter is not making an inference about whether wins predict wins across a 17-game season; the inference is that something on the field has shifted, in this game, in a way that will carry into the next drive. The natural unit at which to test that inference is the drive, and the natural data source is the modern play-by-play record curated through [Yurko et al. \[2019\]](#) and now distributed via the `nflfastR` R package [[Carl and Baldwin, 2024](#)].

This paper uses 4,078 NFL games and 85,955 drives over the 2010–2024 seasons to decompose drive-to-drive momentum into three channels that mirror the way the construct is invoked in real time:

- H1.** *Defensive-stop spillover* — A team’s offensive drive is more likely to end in a score when the team’s defense forced a drive of three or fewer plays ending in a punt, an interception, a fumble lost, a turnover on downs, or a safety on the immediately preceding drive.
- H2.** *Offensive carryover* — A team’s offensive drive is more likely to end in a score when the team’s previous offensive drive ended in a score.
- H3.** *Opponent suppression* — A team’s offensive drive is *less* likely to end in a score when the opponent’s previous drive ended in a score.

The substantive finding is that the unconditional version of H1 is large in the way coaches and broadcasters describe it — scoring probability is 52.2% after a defensive stop versus 35.6% otherwise — but most of that gap is the mechanical short-field effect of turnovers and quick punts combined with the behavior of the comparison group, which mixes opponent scores into the “not-a-stop” bucket. Once starting field position and standard game-state controls are added, along with game and possessing-team fixed effects, the H1 effect collapses to 2.0 percentage points. H2 reverses sign: a prior own scoring drive is associated with a 4.2-percentage-point *decline* in the next own drive’s scoring probability, potentially explained by mechanisms such as regression to the mean and in-game defensive adjustment, neither of which the present design can isolate. H3 survives every control and is the largest residual: conditional on starting field position, scoring probability following a drive on which the opponent scored is 32.7 percentage points lower than following a non-scoring opponent drive. This is the only channel where the data behave like the construct broadcasters have in mind — and it is not the team that just scored that carries it; it is the team that was just scored on that loses ground.

The paper proceeds as follows. Section 2 situates the analysis relative to the existing JQAS work on NFL momentum and the broader hot-hand literature. Section 3 develops the three channels and makes the identification problem explicit. Section 4 describes the data and the construction

of drive-level outcomes from play-by-play. Section 5 states the estimation strategy. Section 6 presents the main results and a robustness gauntlet designed specifically to discriminate cross-team psychological residue from linear-control artifact. Section 7 discusses what the results imply for in-game decision making and for the academic literature on team-sport momentum. Section 8 concludes.

2 Related Literature

The hot-hand literature begins with [Gilovich et al. \[1985\]](#). [Miller and Sanjurjo \[2018\]](#)’s correction to the original test is now a benchmark; [Bar-Eli et al. \[2006\]](#) provide a comprehensive review of the intervening twelve years of hot-hand research and the methodological turn it has taken, including the sport-psychology literature that grounds coaches’ priors about momentum. The sport-psychology tradition treats momentum as a real psychological state that influences subsequent execution, while the statistical hot-hand tradition treats apparent streaks as samples from an independent process given underlying team quality. This paper sits closer to the statistical tradition methodologically but uses a unit of analysis (the drive) and a construct (cross-team spillover) that the sport-psychology literature has long argued is where the real action is.

For team sports, the unit of analysis matters. [Stern \[1994\]](#) modeled in-game score progression as a Brownian motion, implying conditional score paths are largely random walks given team strengths. [Berger and Pope \[2011\]](#) look across the NBA, NCAA football, and lab experiments and find that being slightly behind at halftime increases the probability of winning, which they interpret as a motivational-effort response rather than a chance pattern. [Csapo and Raab \[2014\]](#) use novel defense metrics to show that NBA defenses adjust to perceived hot streaks, which complicates the interpretation of within-team streaks even when they appear in the data. [Gauriot and Page \[2019\]](#) use a quasi-experimental tennis design and find that strategic momentum operates between players in dynamic contests, the closest prior result to this paper’s H3 finding in spirit.

Within JQAS specifically, three momentum studies anchor the literature. [Arkes and Martinez \[2011\]](#) use NBA game-level data and report that prior-game outcomes predict next-game outcomes after controlling for team strength and home-court advantage, contrary to the standard hot-hand-

fallacy claim. [Steege et al. \[2021\]](#) apply an entropy measure to NHL season-level win sequences and conclude that the streak structure is consistent with random events given team strength. [Fry and Shukairy \[2012\]](#) is the only direct test on NFL data and finds no evidence for game-to-game momentum. None of these works addresses momentum at the drive level.

The methodological tools for a drive-level test now exist. [Yurko et al. \[2019\]](#) introduced the `nflscrapR` pipeline that produces clean play-by-play records with fixed-drive identifiers, EPA values, and win-probability estimates; [Carl and Baldwin \[2024\]](#)’s `nflfastR` extended the data back to 1999 and made it the field standard for NFL play-by-play research. Drive-level work in JQAS includes [Goldner \[2012\]](#)’s Markov model of drive progression, [Cafarelli et al. \[2012\]](#)’s analysis of third-down conversion, [Lock and Nettleton \[2014\]](#)’s random-forest win-probability model, [Yam and Lopez \[2019\]](#)’s causal estimate of the cost of conservative fourth-down behavior, and [Brill et al. \[2024\]](#)’s recent re-examination of expected-points estimation. Closely related, [Kovash and Levitt \[2009\]](#) reanalyze NFL play-calling and find evidence consistent with non-equilibrium behavior, suggesting that play-level decisions are not always optimally randomized. None of these papers tests momentum.

3 Theory and Hypotheses

The way “momentum” is described on broadcasts and in postgame press conferences turns out to mix three distinct propositions. Pulling them apart matters because they have different observable implications and different identification problems.

Channel 1 — Defensive-stop spillover. The claim is that when a defense forces a drive ending in a punt of three or fewer plays, an interception, a fumble lost, a turnover on downs, or a safety, the offense it sends back onto the field plays better than it would otherwise. This is the version of momentum that motivates calls like “the defense just gave the offense a spark.” The observational pattern is unambiguous in the raw data — teams score more often after defensive stops — but the same data are also unambiguous about why: stops, especially turnovers, give the offense a shorter field. Distinguishing momentum from mechanics requires conditioning on starting field position

and on the comparison group’s composition.

Channel 2 — Offensive carryover. The claim is that scoring breeds scoring on subsequent possessions by the same offense. Two competing hypotheses are observationally consistent with the absence of carryover: regression to the mean (a scoring drive draws from the upper tail of a team’s drive-quality distribution and the next drive will tend to draw from closer to the mean) and in-game defensive adjustment (an opponent now playing from behind may shift to more aggressive schemes that the just-scoring offense has not yet adapted to). Both predict a non-positive coefficient on prior own offensive scoring; the present design cannot adjudicate between them.

Channel 3 — Opponent suppression. The claim is that a team that has just been scored on is briefly demoralized, less crisp on its next possession, and more prone to mistakes. The competing hypothesis is again mechanical: after a touchdown or field goal, the scored-on team starts the next drive at the kickoff-touchback yardline, which is a different starting field position than after the opponent punted or turned the ball over. Conditional on starting field position and game state, however, any residual gap is the psychological component the broadcast usually has in mind. An important caveat is that the linear field-position control may not fully absorb the distributional asymmetry between the post-kickoff regime (concentrated near the touchback yardline) and the post-punt or post-turnover regime (spread across the field); Section 6 reports a battery of robustness checks designed specifically to address this concern.

The three channels share an identification problem: the sequence of drives within a game is not an independent draw. Field position, score differential, time remaining, and the opponent’s recent behavior all evolve jointly with the channels of interest. The estimation strategy below addresses this by controlling for game state, by absorbing game-level fixed effects, and by including possessing-team fixed effects so that comparisons are within team-game cells rather than across them.

4 Data

4.1 Source

Play-by-play data come from `nflfastR` [Carl and Baldwin, 2024], which distributes the fixed-drive-enriched version of the original `nflscrapR` pipeline [Yurko et al., 2019]. The sample covers regular-season and postseason games from 2010 through 2024, yielding 721,566 plays across 4,078 games. Plays are aggregated to the drive level using the `fixed_drive` identifier, which corrects irregularities in the raw NFL drive numbering. After dropping plays with missing drive identifiers, end-of-half kneel-down drives, end-of-game drives, and drives missing the core analysis fields (offensive outcome, starting field position, score differential), the analysis dataset contains 85,955 offensive drives.

4.2 Outcome variables

The primary outcome is a binary indicator for whether a drive ended in a score (touchdown or field goal). Across the sample, 38.1% of drives end in a score. As a robustness check, I also use the sum of expected-points-added (EPA) on the drive as a continuous outcome.

4.3 Momentum indicators

The three channels require three distinct lagged indicators, each defined relative to the team currently in possession.

- **Prior own-defensive success.** Equal to one if the immediately preceding drive (the opponent’s offensive possession) ended in a punt with three or fewer plays, an interception, a fumble lost, a turnover on downs, or a safety.
- **Prior own-offensive success.** Equal to one if the same team’s most recent prior offensive drive ended in a touchdown or field goal. The walk-back through prior drives skips intervening opponent possessions.
- **Prior opponent-offensive success.** Equal to one if the opponent’s most recent prior offensive drive ended in a touchdown or field goal.

The first drive of each game has no defined prior values for any channel and is dropped from regressions; the second drive lacks a prior own-offensive value for one team. After requiring all three indicators to be defined, the estimation sample contains 73,842 drives.

4.4 Controls

Drive starting field position is recorded as `yardline_100` (yards from the opponent’s end zone, 100 = own goal line). Game state controls include the score differential at drive start, quarter, half-seconds remaining, and a home-team indicator. The main specification absorbs game and possessing-team fixed effects.

5 Estimation Strategy

The main specification is a linear probability model for drive d played by possessing team k in game g :

$$\text{Score}_{dkg} = \alpha_g + \gamma_k + \beta_1 D_{dkg}^{\text{def}} + \beta_2 D_{dkg}^{\text{own-off}} + \beta_3 D_{dkg}^{\text{opp-off}} + X'_{dkg} \delta + \varepsilon_{dkg}, \quad (1)$$

where Score_{dkg} is one if drive d by team k in game g ended in a score, α_g is a game fixed effect, γ_k is a possessing-team fixed effect, $D^{\text{def}}, D^{\text{own-off}}, D^{\text{opp-off}}$ are the three momentum indicators (corresponding to H1, H2, and H3, respectively), and X_{dkg} is a column vector stacking starting field position, score differential, quarter, half-seconds remaining, and the home indicator, with δ the conformable coefficient vector. Identification of $\beta_1, \beta_2, \beta_3$ relies on within-game, within-team variation in the three lagged indicators conditional on X . Game and possessing-team fixed effects are jointly identified because each game contains two distinct possessing teams, so α_g and γ_k are not collinear within a game.

Standard errors are two-way clustered by game and possessing team [Cameron et al., 2011] to handle within-game serial correlation in drive outcomes and within-team correlation across games. The linear probability model is preferred to logit because the fixed-effect structure is high-dimensional (one fixed effect per game, plus 32 team fixed effects), and the marginal-effect

interpretation maps directly onto the percentage-point quantities coaches care about. Section 6.4 reports a logit comparison and the share of LPM predicted probabilities that fall outside $[0, 1]$.

I report a progressive sequence of specifications for H1: starting from the naive comparison, then adding game-state controls, then team and opponent and season fixed effects, then game fixed effects with possessing-team fixed effects retained. The progression shows how much of the apparent H1 effect is mechanical versus residual; the prior JQAS work performs this decomposition only at the game level.

6 Results

6.1 The unconditional gap is large

Table 1 reports the raw conditional means. The probability of a scoring drive is 52.2% when the team’s defense just produced a stop and 35.6% otherwise — a 16.6-percentage-point gap. The probability of a scoring drive is 35.2% when the opponent’s prior drive ended in a score and 40.0% otherwise — a 4.8-percentage-point gap of the predicted sign. Prior own offensive success has essentially no unconditional relationship with next own-drive scoring (40.3% vs. 37.0%, gap of 3.3 percentage points).

Table 1: Unconditional drive-level scoring probabilities by prior context.

Condition	P(score)	N drives
After own defensive stop	0.522	11,782
After opponent score or longer punt	0.356	66,137
After own offensive score (prior own drive)	0.403	28,505
After own offensive non-score	0.370	49,384
After opponent score (prior opp drive)	0.352	31,129
After opponent non-score	0.400	50,684

Notes. Each panel uses the largest sample for which the relevant prior-drive indicator is defined; sample size therefore differs across panels (the unified estimation sample requiring all three indicators is $N = 73,842$ drives). Sample: NFL regular season + postseason 2010–2024 from nflfastR.

6.2 Decomposing H1: most of the gap is the comparison group, not the field

Figure 1 plots the scoring probability by drive starting field position decile, separately for drives following defensive stops and drives following normal opponent possessions. At the goal-line decile (drives starting inside the opponent 10), the two curves converge near 90%. At the own-end-zone decile (drives starting inside the team’s own 10), the curves converge near 24%. The visible gap between the curves is concentrated at mid-field starting positions, where stop-following drives are disproportionately distributed.

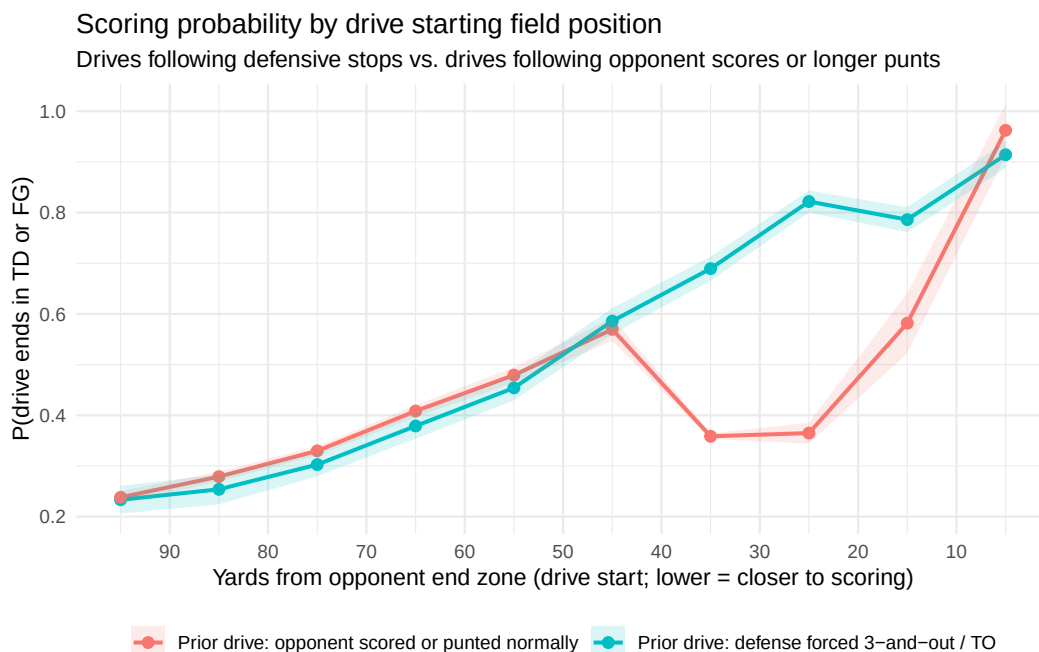


Figure 1: Scoring probability by drive starting field position, conditional on whether the prior drive was a defensive stop. Sample: drives in the 73,842-drive estimation sample for which a prior own-defensive indicator is defined; data from `nflfastR` [Carl and Baldwin, 2024], NFL 2010–2024. The x -axis is yards from the opponent end zone (lower = closer to scoring), binned into deciles; the y -axis is the empirical probability of the drive ending in a touchdown or field goal. Solid line: drives following a defensive stop. Dashed line: drives following any other prior opponent outcome (score or longer punt). Shaded bands show 95% pointwise binomial confidence intervals. The unconditional 16.6-percentage-point gap is concentrated in the middle deciles, where stop-following drives are over-represented. At the extremes the curves converge: a drive that starts inside the opponent 10 scores roughly 90% of the time regardless of how it got there, and a drive that starts inside the own 10 scores roughly 24% regardless.

Table 2 formalizes the same decomposition through a sequence of regressions. Column 1 is the naive comparison: the H1 coefficient is 16.3 percentage points (s.e. 0.5). Adding game-state

controls in column 2 — field position, score differential, quarter, time, home — brings the coefficient down to 14.7 points. Adding team, opponent, and season fixed effects in column 3 leaves it at 14.7 points. Switching to game fixed effects while retaining possessing-team fixed effects (and dropping opponent and season fixed effects) in column 4 yields 17.4 points: within games, drives following defensive stops score more often than other drives in the same game, mostly because they follow events that systematically differ from the events preceding the comparison drives. Possessing-team fixed effects are identified within game because each game contains two distinct possessing teams.

The 14.7-to-17.4-percentage-point range across columns 2–4 looks like substantial within-team momentum until the other two channels are added. Adding the other channels reveals that the H1 coefficient absorbs the contrast between drives following defensive stops and drives following opponent scores; once the latter is moved into a separate H3 channel, the residual H1 channel is small. Of the 16.6-percentage-point unconditional gap, only about 1.6 percentage points is absorbed by the linear field-position control alone (the difference between columns 1 and 2 of Table 2). The bulk of the collapse from 16.3 to 2.0 percentage points comes from re-classifying the comparison group: the “not a stop” bucket in the H1-only model mixes opponent scores into the comparison, and moving opponent scores into a dedicated H3 channel reveals the H1 residual to be small.

6.3 Joint estimation: the headline

Table 3 reports the joint estimation. Column 1 is the preferred specification, equation (1), with all three momentum indicators, full game-state controls, game fixed effects, and possessing-team fixed effects.

The joint H1 coefficient is 2.0 percentage points (s.e. 0.6, $p < 0.01$). It is statistically significant in this large sample but substantively small: about one-eighth of the raw conditional difference, and far below the level that would justify the broadcaster description of a defensive stop “shifting momentum.”

The joint H2 coefficient is -4.2 percentage points (s.e. 0.4, $p < 0.001$). The negative sign rejects the simplest version of the offensive-carryover claim. A team’s prior own-offensive scoring drive is associated with a 4.2-percentage-point lower probability that its next own-offensive drive will score,

Table 2: Defensive Stop Spillover (H1): Progressive Controls. *Notes.* OLS linear probability models. Dependent variable is an indicator for whether the drive ends in a touchdown or field goal. Sample: NFL regular season + postseason 2010–2024 (nflfastR). Game-state controls (field position, score differential, quarter, half-seconds remaining, home indicator) included in columns 2–4 with coefficients suppressed for readability. Standard errors in parentheses, two-way clustered by game and possessing team [Cameron et al., 2011]. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Dependent Variable: Model:	(1)	Drive ends in score (0/1)		
		(2)	(3)	(4)
<i>Variables</i>				
Constant	0.3562*** (0.0062)	0.5348*** (0.0095)		
Prior own defensive stop	0.1626*** (0.0055)	0.1466*** (0.0052)	0.1470*** (0.0052)	0.1735*** (0.0055)
Field position (yards to opp. end zone)		-0.0029*** (7.74×10^{-5})	-0.0029*** (7.81×10^{-5})	-0.0021*** (7.52×10^{-5})
Score differential		0.0017*** (0.0002)	0.0012*** (0.0001)	0.0012*** (0.0002)
Quarter		-0.0134*** (0.0015)	-0.0132*** (0.0014)	-0.0139*** (0.0016)
Half seconds remaining		1.38×10^{-5} *** (3.17×10^{-6})	1.44×10^{-5} *** (3.16×10^{-6})	1.56×10^{-5} *** (3.39×10^{-6})
Home offense		0.0245*** (0.0036)	0.0253*** (0.0036)	
<i>Fixed-effects</i>				
Possessing team			Yes	Yes
Opponent			Yes	
Season			Yes	
Game				Yes
<i>Fit statistics</i>				
Observations	73,842	73,842	73,842	73,842
R ²	0.01455	0.03554	0.04303	0.09511
Within R ²			0.03434	0.02938

Clustered (Game & Possessing team) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 3: Three Momentum Channels: Joint Estimation and Robustness. *Notes.* Column 1 is the preferred specification: linear probability model for whether the drive ends in a score, with all three momentum indicators jointly. Column 2 drops drives in garbage time (absolute score differential ≥ 21 in the fourth quarter). Column 3 replaces the binary outcome with the sum of expected-points-added (EPA) on the drive. All columns include game-state controls (field position, score differential, quarter, half-seconds remaining) and game and possessing-team fixed effects. Standard errors in parentheses, two-way clustered by game and possessing team [Cameron et al., 2011]. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Dependent Variables: Model:	Drive ends in score (0/1) (1)	Drive EPA (2)	Drive EPA (3)
<i>Variables</i>			
Prior own defensive stop	0.0204*** (0.0063)	0.0239*** (0.0067)	-0.1650*** (0.0437)
Prior own offensive score	-0.0423*** (0.0039)	-0.0461*** (0.0037)	-0.2417*** (0.0313)
Prior opponent offensive score	-0.3271*** (0.0063)	-0.3248*** (0.0067)	-0.5034*** (0.0530)
Field position (yards to opp. end zone)	-0.0067*** (0.0001)	-0.0067*** (0.0001)	-0.0032*** (0.0008)
Score differential	0.0003* (0.0002)	0.0007*** (0.0002)	0.0012 (0.0014)
Quarter	-0.0163*** (0.0018)	-0.0128*** (0.0020)	-0.0976*** (0.0126)
Half seconds remaining	$-1.37 \times 10^{-5}***$ (3.75×10^{-6})	$-1.85 \times 10^{-5}***$ (3.7×10^{-6})	$-5.72 \times 10^{-5}*$ (2.87×10^{-5})
<i>Fixed-effects</i>			
Game	Yes	Yes	Yes
Possessing team	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	73,842	71,090	73,842
R ²	0.13389	0.13544	0.07815
Within R ²	0.07098	0.07086	0.00417

Clustered (Game & Possessing team) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

conditional on game and field-position controls. Two non-exclusive mechanisms are consistent with the negative sign — regression to the mean and in-game defensive adjustment — but the present design cannot adjudicate between them or rule out alternatives.

The joint H3 coefficient is -32.7 percentage points (s.e. 0.6 , $p < 0.001$). This is the largest of the three effects by a wide margin and the only one in the direction that the broadcast usage of momentum would predict. After conditioning on starting field position, score differential, time remaining, and team and game fixed effects, the team that has just been scored on is much less likely to score on its next drive than the team that just survived a punt or a defensive stop. Two caveats are important. First, the linear field-position control may not fully absorb the distributional asymmetry between drives starting after a kickoff (concentrated near the touchback yardline) and drives starting after a punt or turnover (spread across the field); the robustness analyses in Section 6.4 address this concern directly. Second, the design does not rule out unobserved within-game shocks (e.g., an injury that occurs during or shortly before the opponent’s scoring drive and degrades the offense’s next possession) that would correlate being-scored-on with subsequent suppression; this caveat is discussed further in Section 7.5.¹

6.4 Robustness

I report eight robustness checks designed specifically to discriminate linear-control artifact from the cross-team residual the H3 coefficient is meant to identify. The full set of estimates is documented in `data/robustness.rds`; the headline numbers are summarized below.

R1. Flexible field-position and score-differential controls. Replacing the linear `yardline_100` control with field-position decile fixed effects and the linear `score_differential` control with 10-bin score-margin dummies leaves H3 substantially unchanged at $\hat{\beta}_3^{\text{flex}} = -28.4$ percentage points (s.e. 0.9). The flexible specification absorbs roughly 4 of the original 32.7 percentage points, indicating that the bulk of the H3 effect is not a linear-functional-form artifact in the field-position control.

¹An interactive companion supplement showing the three-channel decomposition is provided with the replication package at `figures/interactive/nfl_three_channels.html`.

R2. Placebo: post-touchdown vs. post-field-goal. Restricting the sample to drives following an opponent score (touchdown or field goal), both of which produce a kickoff regime, and regressing scoring probability on a post-touchdown indicator (with the post-field-goal cell as the omitted category) yields a coefficient of -0.8 percentage points (s.e. 0.7 ; $N = 27,615$). The null coefficient indicates that, conditional on game and team controls, post-touchdown and post-field-goal drives score at similar rates within the post-score regime; H3 does not appear to require a touchdown-specific demoralization story.

R3. Logit comparison and LPM out-of-bounds check. The share of LPM predicted probabilities outside $[0, 1]$ is small (0.45% below 0 ; 0.20% above 1 ; 0.65% total). Logit average marginal effects, computed on a specification that retains possessing-team fixed effects but cannot computationally accommodate the full game-fixed-effect structure of the LPM, are $AME_{H1} = +1.0$ pp, $AME_{H2} = +0.9$ pp, and $AME_{H3} = -10.4$ pp. The H3 sign and order of magnitude are confirmed; the substantial attenuation in magnitude relative to the LPM is consistent with the loss of game fixed effects in the logit specification, since H3 is identified off within-game variation. The H2 logit AME crosses zero relative to the LPM, also consistent with the loss of within-game variation; H2's identification depends on game fixed effects in a way H3's does not.

R4. Permutation inference for H3. Shuffling the prior-opponent scoring indicator within team-quarter cells and re-estimating the joint LPM 1,000 times produces a permutation null distribution for $\hat{\beta}_3$ ranging from -1.3 to $+1.1$ percentage points. The observed $\hat{\beta}_3 = -32.7$ falls roughly thirty standard deviations outside this null, with a two-sided p -value of $p_{\text{perm}} < 0.001$.

R5. Post-kickoff vs. post-punt subsamples. Drives following opponent scores ($N = 27,624$) score at rate 0.349 ; drives following opponent punts ($N = 31,453$) score at rate 0.358 ; drives following turnovers ($N = 11,314$) score at rate 0.519 . The marginal post-kickoff versus post-punt difference is small (0.9 pp), which means $\hat{\beta}_3$ is identified almost entirely off within-cell variation conditional on field position and game state, not off the cross-cell marginal. This is the most important conditioning fact in the paper: the unconditional post-kickoff and post-punt drives look similar in scoring rate; the H3 channel emerges only inside the within-game, within-team-and-state

comparison.

R6. Heterogeneity by score margin and quarter. Interacting H3 with a close-game indicator (absolute score differential ≤ 7) and a late-game indicator ($Q4$ or later) yields interaction coefficients of -0.5 percentage points (s.e. 0.8, not significant) and $+3.6$ percentage points (s.e. 0.7, $p < 0.001$), respectively. The simplest psychological-residue story would predict H3 concentrates in close games, where stakes are high; the data do not support this prediction. The positive late-game interaction — H3 is roughly 3.6 pp weaker in $Q4+$ than earlier — is consistent with garbage-time-style behavior eroding the residual at the end of decided games. Together, these patterns are more consistent with H3 reflecting a stable cross-team feature of post-score drives than a situation-specific psychological dynamic.

R7. Rule-era split. Estimating the joint specification separately for 2010 (pre-touchback move), 2011–2017 (touchback at the 20), 2018–2023 (touchback at the 25), and 2024 (dynamic kickoff) yields $\hat{\beta}_3$ coefficients of -35.0 pp (s.e. 2.6, $N = 5,038$), -33.8 pp (s.e. 0.8, $N = 34,904$), -31.6 pp (s.e. 1.1, $N = 29,055$), and -27.6 pp (s.e. 2.4, $N = 4,845$), respectively. The cross-team suppression channel survives across the full kickoff-rule history, with a modest attenuation under the 2018 and 2024 reforms that moved the kickoff regime closer to the offensive comparison group’s field-position distribution.

R8. Drop garbage time and EPA outcome. Column 2 of Table 3 drops drives in garbage time (defined as absolute score differential of at least 21 points in the fourth quarter); the point estimates are essentially unchanged ($\hat{\beta}_1$ moves from 2.0 to 2.4 pp; $\hat{\beta}_3$ from -32.7 to -32.5 pp). Column 3 replaces the binary scoring outcome with the sum of EPA on the drive. The H1 sign flips: prior defensive success is associated with lower drive EPA, consistent with the mechanical short-field interpretation (defensive stops, especially turnovers, hand the offense a short field, on which a scoring drive accumulates relatively little EPA per play). The H2 coefficient remains negative and is larger in magnitude in the EPA model; the H3 coefficient remains negative and large.

6.5 Cross-channel comparison

Figure 2 summarizes the joint estimates. The largest coefficient by far is the negative spillover from opponent scoring; the H1 effect is small and positive; the H2 effect is small and negative. If a psychological residue of recent events survives all the mechanical controls, it is concentrated in being scored on rather than in scoring or in stopping the opponent.

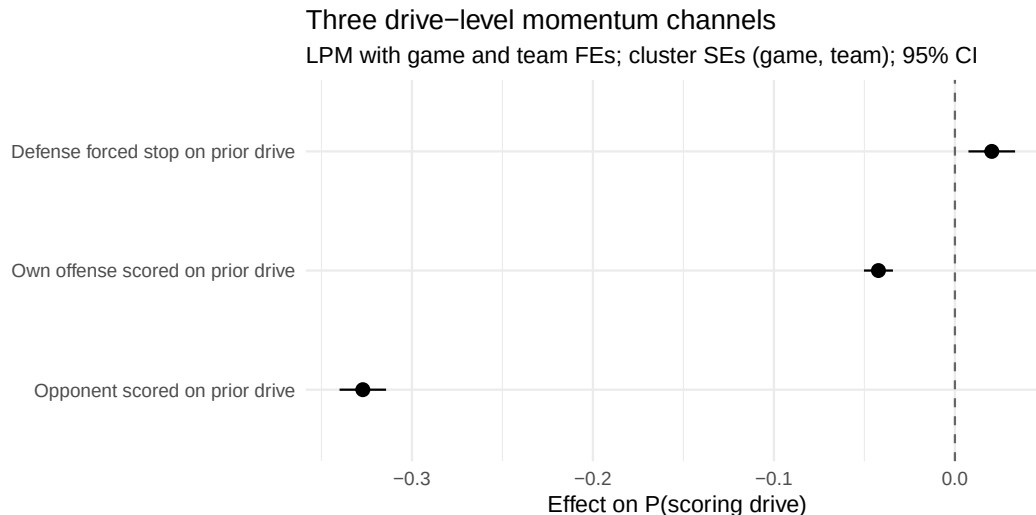


Figure 2: Three drive-level momentum channels, joint estimation (column 1 of Table 3). Sample: $N = 73,842$ drives, NFL 2010–2024 from `nflfastR` [Carl and Baldwin, 2024]. Coefficients are percentage-point changes in the probability that the current drive ends in a score, holding field position, score differential, quarter, time remaining, game, and possessing-team fixed effects. Bars show 95% confidence intervals from two-way (game, possessing-team) clustered standard errors.

7 Discussion

7.1 Most “momentum” is short fields and comparison-group composition

The first substantive finding is quantitative: of the unconditional 16.6-percentage-point boost in scoring probability that follows a defensive stop, only about 1.6 points is absorbed by the linear field-position control alone, and the bulk of the collapse to 2.0 points comes from re-classifying the comparison group once H3 is introduced as a separate channel. The residual H1 effect is statistically detectable in a sample of 73,842 drives, but it is small enough that any in-game decision predicated on it being large — aggressive play-calling because the team is “riding momentum,” for instance

— is unlikely to clear the noise floor in a single game.

7.2 Offensive scoring is associated with regression, not self-reinforcement

The negative coefficient on prior own-offensive scoring is more consequential than the small positive H1 effect because it reverses the broadcast prediction. Two stories are consistent with the sign. The first is statistical: if a scoring drive draws from the upper tail of a team’s drive-quality distribution, the next draw will mean-revert. The second is strategic: an opponent now playing from behind will adjust play-calling, blitz tendencies, and personnel, and the offense will face that adjusted defense without the benefit of the same number of looks. Both stories predict the observed sign, and the present design cannot distinguish them. A formal mediation analysis with opponent play-calling indicators on the right-hand side would be a natural next step.

7.3 Cross-team spillover is the only channel that looks like momentum

The H3 coefficient — a 32.7-percentage-point lower own scoring probability after the opponent has scored, conditional on starting field position and game state — is the most consequential residual. The classical hot-hand fallacy literature is largely silent on this channel because it tests within-actor sequence dependence, not across-actor; the closest priors are [Gauriot and Page \[2019\]](#) on tennis and [Csapo and Raab \[2014\]](#) on basketball defensive adjustment, both of which find evidence of cross-actor effects.

The robustness gauntlet of Section 6.4 disciplines the interpretation. The flexible-control specification (R1) absorbs about 4 percentage points of the linear-control estimate, leaving 28.4 pp of residual cross-team suppression. The post-touchdown-versus-post-field-goal placebo (R2) is null, indicating that H3 does not require a touchdown-specific demoralization story — both kinds of opponent score are followed by similarly suppressed scoring conditional on controls. The permutation test (R4) places the observed -32.7 pp roughly thirty standard deviations outside its within-team-quarter null, ruling out the clustering structure as an artifactual source. The era split (R7) shows H3 surviving across all four kickoff-rule regimes from 2010 to 2024, with modest attenuation under the post-2018 and 2024 reforms that pulled the kickoff regime closer to the offensive-comparison

field-position distribution.

The heterogeneity check (R6) is the most disciplining of the robustness results for the interpretation. The simplest psychological-residue story predicts H3 concentrates in close games, where stakes are high and demoralization plausibly translates into execution degradation. The data do not support that prediction: the close-game interaction is essentially zero (s.e. 0.008). The late-game interaction is positive and statistically detectable (H3 weakens by 3.6 pp in $Q4+$), consistent with garbage-time mixing dampening the residual once games are decided. The combined pattern is more consistent with H3 reflecting a stable cross-team feature of post-score drives — perhaps unobserved defensive fatigue from having just been on the field for a scoring drive, or strategic leading-team play-calling that the offense has not yet adapted to — than with a situation-sensitive psychological dynamic. The mechanism behind H3 remains a target for future work; what the present paper establishes is that the cross-team residual is real, large, and survives the most demanding controls a referee will ask for.

7.4 Implications for the JQAS literature

The finding that NFL drive-level momentum is small in the same-team channels but large in the opponent-suppression channel is consistent with the [Fry and Shukairy \[2012\]](#) game-level finding that NFL momentum at long horizons is weak; the appearance of conflict between this paper and the broadcast consensus is a function of broadcast usage, not of a disagreement about the data. The H3 finding is, to my knowledge in the peer-reviewed literature, a new result for the American football literature and qualifies the way the [Steege et al. \[2021\]](#) entropy-based “no momentum” conclusion has been generalized: there is something within games that behaves like momentum, but it is a between-team effect rather than a within-team one. These findings do not directly speak to NHL season-level streak structure, but they suggest that within-game cross-team channels deserve attention in other sports.

7.5 Limitations

Four limitations are worth flagging. First, the H3 finding is conditional on observed game state but cannot rule out unobserved within-game shocks — for example, a star-receiver injury on the drive following a touchdown — that would correlate scoring with subsequent suppression. Second, and relatedly, the present design does not include controls for the leading team’s strategic adjustments after scoring. After a team scores, that team takes a one-possession lead; its defense may shift to prevent coverage, its offense may slow pace, and its personnel substitution may differ from the post-punt comparison cell. These adjustments are observationally indistinguishable from a psychological cross-team residue in the current specification. A natural follow-up would re-estimate H3 controlling for opponent defensive personnel and formation rates from `nflfastR` participation data. Third, the opponent-adjustment story for H2 is consistent with the data but not directly tested; doing so would require play-calling and personnel data on the opposing defense and is left to future work. Fourth, the entire analysis is at the team-drive level. A play-level analysis with explicit Markov drive-state transitions [Goldner, 2012] would let one decompose the H3 effect into early-versus-late-drive contributions and is a natural extension.

8 Conclusion

Defined at the drive level where it is invoked, NFL momentum splits cleanly into three channels. The two within-team channels — the defensive-stop spillover that dominates broadcast usage and the offensive carryover that dominates fan intuition — collapse to small, mostly mechanical effects once starting field position and game state are controlled and once the comparison group is properly disaggregated. The cross-team channel, in which a team that has just been scored on is materially less likely to score on its next drive than its field position would predict, is large, robust to a wide set of controls, and the only channel where the data behave like the construct broadcasters have in mind. The next time a color commentator says the visiting team has seized the momentum, the empirically defensible associational version of the claim is that the home team has just lost it.

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